COUNCIL OF OPERATIONAL KILL-CHAIN EXPERIMENTATION & DEMONSTRATION (COOKED)

Denver, CO

TURBODRONE: Tactical Unmanned Reconnaissance Battle Operations Detecting Rogue OppoNEnts

STATEMENT OF WORK (SOW)

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Revision History

FINAL_REV3

Ground-Based Defense Network Requirements

1.8 Tracking Targets
The Ground-Based Defense Network shall determine the zone (outer, middle, or central) of any singular rogue aerial target that is moving at ≤ 1 m/s, with a positional accuracy of ± 5 meters, and shall transmit this zone-membership information to the Forward Command Post within 2 seconds of detection.

FINAL_REV2

Ground-Based Defense Network Requirements

1.1 Protected Area Layout
The protected area shall be organized into 3 simultaneously operating quarter-circular zones. Each zone shall extend outward from a common center point covering a ground area of 90 degrees and an airspace altitude range of 0-15m measured from ground level.

1.2 Zone Layout
The zones in the protected area shall be an outer zone with a radius of 85m for detection, a middle zone with a radius of 60m for tracking, and a central zone with a radius of 30m for engagement.

Operations Analysis Requirements

4.7 Protected Areas
The second OA study shall model and visualize the enclosure volume of 3 protected areas of various sizes between 500-1500km diameter and 0-2000km altitude.

4.2 Proposal Trade Study Deliverables
The initial OA study shall be incorporated into the proposal in the form of a whitepaper without the development of any digital models, scenarios, or data analysis.

4.5 Demo Trade Study
A second, independent, OA study shall design, model, and simulate a remote-sensing constellation detecting a cross-domain adversarial network of targets including surface vessels, aircraft, ballistic missiles, and satellites entering the protected areas during the mission scenario. Each target in the adversarial network shall exhibit a phenomenology that allows it to be detected by the remote-sensing constellation. The second OA study may use all, some, or none of the

research from the proposal trade study to guide decision-making during the demo trade study.

FINAL_REV1

Demonstration Mission - Thread 1: Ground-Based Defense Network

1 Enclose three zones that originate from a common central point

FINAL

Ground-Based Defense Network Requirements

1.1 Protected Area Layout
The protected area shall be organized into 3 simultaneously operating semi-circular zones. Each zone shall extend outward from a common center point covering a ground area of 0-180 degrees and an airspace altitude range of 0-15m measured from ground level.

1.2 Zone Layout
The zones in the protected area shall be an outer zone with a radius of 90m for detection, a middle zone with a radius of 60m for tracking, and a central zone with a radius of 30m for engagement.

1.3 Network Design
The Ground-Based Defense Network shall be composed of a distributed set of nodes providing total coverage of the protected area. At least one mode is expected of being used at all times in each zone. Each mode shall be capable of operating independently of the others, while also cooperating together when multiple sensors are active

1.4 Field Data Relay
The Ground-Based Defense Network shall relay field data back to the forward command post for real-time visualization and mission-processing.

1.8 Tracking Targets
The Ground-Based Defense Network shall determine the zone (outer, middle, or central) of any singular rogue aerial target that is moving at ≤ 1 m/s, with a positional accuracy of ± 5 meters, and shall transmit this zone-membership information to the Forward Command Post within 2 seconds of detection.

1.9 Engaging Targets
The Ground-Based Defense Network shall demonstrate a visual confirmation that an engagement was executed against any rogue aerial target moving at a speed below 3 m/s that enters the central zone.

1.10 Engagement Safety
The Ground-Based Defense Network shall execute all engagements in a way that’s incapable of causing bodily injury or damage to any non-rogue aerial targets.

Forward Command Post Requirements

3.1 Real-Time Video Feed
The Forward Command Post Shall be capable of displaying a video stream (at least 30 fps) of targets being detected and tracked by the Ground-Based Defense Network.

3.2 Computer Based Visualization
The Forward Operating Command Post shall use computer vision to display real-time (at least 30 fps) target tracking confidence levels during mission operations.

3.4 Node Network Status
The Forward Command Post shall display a low-fidelity network map showing the health, status, engagement area, rough location of nodes in the engagement area, forward command post, and network links of the nodes in the Ground-Based Defense Network at a maximum latency of 1000 milliseconds.

3.5 Alerts and Notifications
The Forward Command Post shall give the user alerts that indicate fault or failure, and notifications are information or change in state/status of the system for node network for issues related to signal and power levels at a maximum latency of 5000 milliseconds.

Operations Analysis Requirements

4.2 Proposal Trade Study Deliverable
The second OA study shall be incorporated into the proposal in the form of a whitepaper without the development of any digital models, scenarios, or data analysis.

Safety Requirements

6.1 Electrical Discharge
All hardware delivered or used by the contractor shall prevent electrical discharge to system handlers and be compliant with MIL-STD-1472G Section 5.7.

Introduction

The multi-domain protection of geographical regions from remote threats is a critical area of technology development. The dynamic and evolving nature of these threats demands a proactive and innovative approach to surveillance and monitoring, one that can provide real-time situational awareness and enable swift decision-making.

To address this imperative, COOKED solicits the development of a distributed ground-based defense network to find, fix, track, target, engage, and assess (F2T2EA) rogue aerial targets. It will also enable the integration and testing of various subsystems, including sensors, command and control systems, and effectors. The ground network will be enhanced by the integration of artificial intelligence (AI) for the autonomous handling of F2T2EA events during the course of the mission.

To facilitate the engagement between the ground and air domains, rogue aerial targets will maneuver within an enclosed test area that's divided into 3 different semi-circular zones. Each zone will contain nodes equipped with different sensors – including optical, RF, acoustic, and ultrasonic – to handle F2T2EA events. Should any target reach the central zone, the defense network shall demonstrate a visual confirmation that an engagement was executed against that target.

Additionally, all field data generated during the mission from the ground network will be relayed in real-time to a forward command post, providing a comprehensive situational awareness picture including a visualization of the autonomous tracking and status of the ground network. The autonomous tracking status is indicated by a series of visual cues showing metrics such as the number of targets being tracked, their location, and the confidence level of the tracking. Meanwhile, the network status is displayed on a digital layout containing a series of nodes and links with indicators showing the health and connectivity of each sensor and communication pathway.

Finally, to inform the development of this capability in the space domain, COOKED solicits an Operations Analysis (OA) study to investigate the design, modeling, and simulation of a distributed remote sensing constellation that can provide persistent, multi-domain surveillance and monitoring of designated protected areas. The OA study will focus on developing a system capable of detecting and tracking phenomenology-based objects, with the goal of informing the development of a modular, flexible, and scalable architecture for future missions.

Demonstration Mission

The goal of TURBODRONE is to demonstrate the key technologies needed for a ground-based defense network to automatically detect, track, and engage a network of rogue aerial targets.

The prime contractor will perform a demonstration mission between 0900 and 1200 hours local time, and be conducted at the employee ballfields at Waterton Campus (Littleton). The demonstration mission includes the following mission threads to be presented and/or executed:

Thread 1: Ground-Based Defense Network

1. Enclose three zones that originate from a common central point
2. Autonomously detect, track, and engage aerial targets
3. Relay field data to forward command post

Thread 2: Rogue Aerial Targets

1. Sustain at least a 10min flight and demonstrate all capabilities of Ground-Based Defense Network
2. Maneuver in and out of protected area airspace

Thread 3: Forward Command Post Operations

1. Real-time video live stream of detections and tracking events
2. Visualize status of tracking with computer vision
3. Display the health, status, and location of each node in the network

Thread 4: Operations Analysis Presentation

1. Proposal Trade Study - Explore the characteristics of different phenomenologies in space
2. Demo Trade Study – Design, model, and simulate a remote-sensing constellation capable of detecting objects entering protected areas

Requirements

1 Ground-Based Defense Network Requirements

1.1 Protected Area Layout
The protected area shall be organized into 3 simultaneously operating quarter-circular zones. Each zone shall extend outward from a common center point covering a ground area of 90 degrees and an airspace altitude range of 0-15m measured from ground level.

1.2 Zone Layout
The zones in the protected area shall be an outer zone with a radius of 85m for detection, a middle zone with a radius of 60m for tracking, and a central zone with a radius of 30m for engagement.

1.3 Network Design
The Ground-Based Defense Network shall be composed of a distributed set of nodes providing total coverage of the protected area. At least one mode is expected of being used at all times in each zone. Each mode shall be capable of operating independently of the others, while also cooperating together when multiple sensors are active.

1.4 Field Data Relay
The Ground-Based Defense Network shall relay field data back to the forward command post for real-time visualization and mission-processing.

1.5 Detection Modes
The Ground-Based Defense Network shall be capable of supporting multiple detection modes within each zone of the network during operations - including optical, passive RF, acoustic, and ultrasonic.

1.6 Detection Mode Selection
The Ground-Based Defense Network shall be able to receive commands from the forward command post to update detection modes being used within each zone of the protected area.

1.7 Detecting Targets
The Ground-Based Defense Network shall report the latitude, longitude, and altitude (LLA) coordinates of any singular rogue aerial target moving below 5 m/s that enters the outer zone of the protected area back to the forward command post.

1.8 Tracking Targets
The Ground-Based Defense Network shall determine and report the zone membership (outer/middle/central) of any singular rogue aerial target

moving below 5 m/s. Zone membership updates shall be reported to the Forward Command Post within 2000 milliseconds of a zone membership.

1.9 Engaging Targets
The Ground-Based Defense Network shall demonstrate a visual confirmation that an engagement was executed against any rogue aerial target moving at a speed below 3 m/s that enters the central zone.

1.10 Engagement Safety
The Ground-Based Defense Network shall execute all engagements in a way that's incapable of causing bodily injury or damage to any non-rogue aerial targets.

1.11 Primary Power
The Ground-Based Defense Network shall operate on single phase 120 VAC 60 Hz electrical power or be self-powered for the duration of the mission.

1.12 Backup Power
The Ground-Based Defense Network shall demonstrate the ability of solar energy as an alternative remote power source.

2 Rogue Aerial Target Network Requirements

2.1 Airframe
Roque Aerial Targets shall be rotor-propelled aircraft capable of stable flight within the enclosed protected area.

2.2 Endurance
Roque Aerial Targets shall be capable of a minimum continuous flight time of 10 minutes under nominal payload and environmental conditions.

2.3 Power
Roque Aerial Targets shall be self-powered for the duration of the mission.

2.4 Maintainability
Roque Aerial Targets shall allow field replacement of propellers and batteries within 10 minutes by a trained operator.

2.5 Safety & FAA Compliance
Roque Aerial Targets shall comply with FAA Part 107 small UAS requirements.

3 Forward Command Post Requirements

3.1 Real–Time Video Feed
The Forward Command Post Shall be capable of displaying a video stream (at least 30 fps) of targets being detected and tracked by the Ground-Based Defense Network.

3.2 Computer Based Visualization
The Forward Operating Command Post shall use computer vision to display real-time (at least 30 fps) target tracking confidence levels during mission operations.

3.3 F2T2EA Performance Analysis
The Forward Command Post shall display an analysis of F2T2EA events and their rates, durations, and losses for each target, and display the following performance metrics post mission-operations: identification accuracy and engagement success rates, sensor mode usages and durations, and the average time from detection to engagement for each target.

3.4 Node Network Status
The Forward Command Post shall display a low-fidelity network map showing the health, status, engagement area, rough location of nodes in the engagement area, forward command post, and network links of the nodes in the Ground-Based Defense Network at a maximum latency of 1000 milliseconds.

3.5 Alerts and Notifications
The Forward Command Post shall give the user alerts that indicate fault or failure, and notifications are information or change in state/status of the system for node network for issues related to signal and power levels at a maximum latency of 5000 milliseconds.

3.6 Power
The Forward Command Post shall operate on single phase 120 VAC 60 Hz electrical power or be self-powered for the duration of the mission.

3.7 Ground-Based Defense Network Commanding
The Forward Command Post shall be capable of sending commands to the Ground-Based Defense Network over secure communications links.

4 Operations Analysis Requirements

4.1 Proposal Trade Study
An initial OA study that explores electromagnetic, space weather, radiation, and optical phenomenologies shall be included as part of the contractor’s proposal submission for TURBODRONE.

4.2 Proposal Trade Study Deliverables
The initial OA study shall be incorporated into the proposal in the form of a whitepaper without the development of any digital models, scenarios, or data analysis.

4.3 Phenomenology Comparison
The initial OA study shall compare and contrast the scientific and physical characteristics of each phenomenology of interest.

4.4 Resource Comparison
The initial OA study shall compare the kinds of resources that are needed to detect each phenomenology of interest.

4.5 Demo Trade Study
A second, independent, OA study shall design, model, and simulate a remote-sensing constellation detecting a cross-domain adversarial network of targets including surface vessels, aircraft, ballistic missiles, and satellites entering the protected areas during the mission scenario. Each target in the adversarial network shall exhibit a phenomenology that allows it to be detected by the remote-sensing constellation. The second OA study may use all, some, or none of the research from the proposal trade study to guide decision-making during the demo trade study.

4.6 Demo Trade Study Deliverables
The second OA study shall deliver a research product that includes a digital model of the mission scenario and access to any code developed for data analysis.

4.7 Protected Areas
The second OA study shall model and visualize the enclosure volume of 3 protected areas of various sizes between 500-1500km diameter and 0-1500km altitude.

4.8 Protected Area Coverage
The second OA study shall be capable of monitoring the protected areas 99.5% of time over the course of 30 days.

4.9 Ground Station Network
The second OA study shall include multiple ground terminals capable of receiving downlinks from the remote-sensing constellation at a minimum frequency of every 90-minutes.

4.10 Network Visualization
The second OA study shall visualize links between the remote-sensing constellation, targets, and ground terminals.

4.11 Coverage Analysis
The second OA study shall calculate the number of access windows, gap windows, and the distribution in lengths of coverage in the constellation to determine how many minutes each ground station remains within contact over a period of 30 days.

4.12 Communication Analysis
The second OA study shall calculate and model the optimal data rate and link budget of the remote-sensing network.

4.13 Cost Analysis
The second OA study shall estimate the cost in USD to develop, launch, populate, and operate the remote-sensing constellation based on historical trends of similar past missions.

4.14 Schedule Analysis
The second OA study shall estimate the schedule needed to develop, launch, populate, and operate the remote-sensing constellation based on historical trends of similar past missions.

5 Demonstration Event Requirements

5.1 Mission Setup
The contractor shall setup all demo equipment prior to 0900 hours local time on the day of demonstration.

5.2 Mission Start
The contractor shall commence execution of the mission at 0900 hours local time.

5.3 Mission Completion
The contractor shall complete execution of all mission threads specified in the TURBODRONE SOW by 1200 hours local time.

5.4 Mission Presentation
The contractor shall present to COOKED representatives throughout mission execution. COOKED will provide the minimum required presentation contents to the contractor no later than 1 month prior to demonstration mission execution.

5.5 OA Study Presentation
The contractor shall present the design of the mission scenario, the process of conducting the OA study, a demonstration of the mission scenario running in the simulation environment, and a conclusive report of its findings.

6 Safety Requirements

6.1 Electrical Discharge

All hardware delivered or used by the contractor shall prevent electrical discharge to system handlers and be compliant with MIL-STD-1472G Section 5.7.

6.2 Safety Inspection

All hardware delivered or used by the contractor shall pass a customer safety inspection on demonstration day prior to mission execution.

6.3 Safety Labels

All hardware shall possess safety labels and markings per MIL-STD-1472G Section 5.7, display of warnings and hazards.

6.4 Safety Training

All contractor members shall complete Atlas Learning Course: Fire Safety and Prevention 2.0 (ehs_hsf_a93_sh_enus) prior to start of assembly and testing.

6.5 Training Manual

The contractor shall provide a training manual for system operation and handling procedures.